

RAG-ANYTHING: ALL-IN-ONE RAG FRAMEWORK

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ABSTRACT

Retrieval-Augmented Generation (RAG) has emerged as a fundamental paradigm for expanding Large Language Models beyond their static training limitations. However, a critical misalignment exists between current RAG capabilities and real-world information environments. Modern knowledge repositories are inherently multimodal, containing rich combinations of textual content, visual elements, structured tables, and mathematical expressions. Yet existing RAG frameworks are limited to textual content, creating fundamental gaps when processing multimodal documents. We present RAG-Anything, a unified framework that enables comprehensive knowledge retrieval across all modalities. Our approach reconceptualizes multimodal content as interconnected knowledge entities rather than isolated data types. The framework introduces dual-graph construction to capture both cross-modal relationships and textual semantics within a unified representation. We develop cross-modal hybrid retrieval that combines structural knowledge navigation with semantic matching. This enables effective reasoning over heterogeneous content where relevant evidence spans multiple modalities. RAG-Anything demonstrates superior performance on challenging multimodal benchmarks, achieving significant improvements over state-of-the-art methods. Performance gains become particularly pronounced on long documents where traditional approaches fail. Our framework establishes a new paradigm for multimodal knowledge access, eliminating the architectural fragmentation that constrains current systems. Our framework is open-sourced at: <https://github.com/HKUDS/RAG-Anything>.

1 INTRODUCTION

Retrieval-Augmented Generation (RAG) has emerged as a fundamental paradigm for expanding the knowledge boundaries of Large Language Models (LLM) beyond their static training limitations [Zhang et al. \(2025\)](#). By enabling dynamic retrieval and incorporation of external knowledge during inference, RAG systems transform static language models into adaptive, knowledge-aware systems. This capability has proven essential for applications requiring up-to-date information, domain-specific knowledge, or factual grounding that extends beyond pre-training corpora.

However, existing RAG frameworks focus exclusively on text-only knowledge while neglecting the rich multimodal information present in real-world documents. This limitation fundamentally misaligns with how information exists in authentic environments. Real-world knowledge repositories are inherently heterogeneous and multimodal [Abootorabi et al. \(2025\)](#). They contain rich combinations of textual content, visual elements, structured tables, and mathematical expressions across diverse document formats. This textual assumption forces existing RAG systems to either discard non-textual information entirely or flatten complex multimodal content into inadequate textual approximations.

The consequences of this limitation become particularly severe in document-intensive domains where multimodal content carries essential meaning. Academic research, financial analysis, and technical documentation represent prime examples of knowledge-rich environments. These domains fundamentally depend on visual and structured information. Critical insights are often encoded exclusively in non-textual formats. Such formats resist meaningful conversion to plain text.

The consequences of this limitation become particularly severe in knowledge-intensive domains where multimodal content carries essential meaning. Three representative scenarios illustrate the critical

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need for multimodal RAG capabilities. In **Scientific Research**, experimental results are primarily communicated through plots, diagrams, and statistical visualizations. These contain core discoveries that remain invisible to text-only systems. **Financial Analysis** relies heavily on market charts, correlation matrices, and performance tables. Investment insights are encoded in visual patterns rather than textual descriptions. Additionally, **Medical Literature Analysis** depends on radiological images, diagnostic charts, and clinical data tables. These contain life-critical information essential for accurate diagnosis and treatment decisions. Current RAG frameworks systematically exclude these vital knowledge sources across all three scenarios. This creates fundamental gaps that render them inadequate for real-world applications requiring comprehensive information understanding. Therefore, multimodal RAG emerges as a critical advancement. It is necessary to bridge these knowledge gaps and enable truly comprehensive intelligence across all modalities of human knowledge representation.

Addressing multimodal RAG presents three fundamental technical challenges that demand principled solutions. This makes it significantly more complex than traditional text-only approaches. The naive solution of converting all multimodal content to textual descriptions introduces severe information loss. Visual elements such as charts, diagrams, and spatial layouts contain semantic richness that cannot be adequately captured through text alone. These inherent limitations necessitate the design of effective technical components. Such components must be specifically designed to handle multimodal complexity and preserve the full spectrum of information contained within diverse content types.

Technical Challenges. • **First**, the **unified multimodal representation** challenge requires seamlessly integrating diverse information types. The system must preserve their unique characteristics and cross-modal relationships. This demands advanced multimodal encoders that can capture both intra-modal and inter-modal dependencies without losing essential visual semantics. • **Second**, the **structure-aware decomposition** challenge demands intelligent parsing of complex layouts. The system must maintain spatial and hierarchical relationships crucial for understanding. This requires specialized layout-aware parsing modules that can interpret document structure and preserve contextual positioning of multimodal elements. • **Third**, the **cross-modal retrieval** challenge necessitates sophisticated mechanisms that can navigate between different modalities. These mechanisms must reason over their interconnections during retrieval. This calls for cross-modal alignment systems capable of understanding semantic correspondences across text, images, and structured data. These challenges are amplified in long-context scenarios. Relevant evidence is dispersed across multiple modalities and sections, requiring coordinated reasoning across heterogeneous information sources.

Our Contributions. To address these challenges, we introduce RAG-Anything, a unified framework that fundamentally reimagines multimodal knowledge representation and retrieval. Our approach employs a **dual-graph construction strategy** that elegantly bridges the gap between cross-modal understanding and fine-grained textual semantics. Rather than forcing diverse modalities into text-centric pipelines, RAG-Anything constructs **complementary knowledge graphs** that preserve both multimodal contextual relationships and detailed textual knowledge. This design enables seamless integration of visual elements, structured data, and mathematical expressions within a unified retrieval framework. The system maintains **semantic integrity** across modalities while ensuring efficient **cross-modal reasoning capabilities** throughout the process.

Our **cross-modal hybrid retrieval** mechanism strategically combines **structural knowledge navigation** with **semantic similarity matching**. This architecture addresses the fundamental limitation of existing approaches that rely solely on embedding-based retrieval or keyword matching. RAG-Anything leverages explicit graph relationships to capture multi-hop reasoning patterns. It simultaneously employs dense vector representations to identify semantically relevant content that lacks direct structural connections. The framework introduces **modality-aware query processing** and **cross-modal alignment systems**. These enable textual queries to effectively access visual and structured information. This unified approach eliminates the architectural fragmentation that plagues current multimodal RAG systems. It delivers superior performance particularly on long-context documents where relevant evidence spans multiple modalities and document sections.

Experimental Validation. To validate the effectiveness of our proposed approach, we conduct comprehensive experiments on two challenging multimodal benchmarks: DocBench and MMLongBench. Our evaluation demonstrates that RAG-Anything achieves superior performance across diverse domains. The framework represents substantial improvements over state-of-the-art baselines. Notably, our performance gains become increasingly significant as content length increases. We observe particularly pronounced advantages on long-context materials. This validates our core hypothesis

that dual-graph construction and cross-modal hybrid retrieval are essential for handling complex multimodal materials. Our ablation studies reveal that graph-based knowledge representation provides the primary performance gains. Traditional chunk-based approaches fail to capture the structural relationships critical for multimodal reasoning. Case studies further demonstrate that our framework excels at precise localization within complex layouts. The system effectively disambiguates similar terminology and navigates multi-panel visualizations through structure-aware retrieval mechanisms.

2 THE RAG-ANYTHING FRAMEWORK

2.1 PRELIMINARY

Retrieval-Augmented Generation (RAG) has emerged as a fundamental paradigm for dynamically expanding the knowledge boundaries of LLMs. While LLMs demonstrate exceptional reasoning capabilities, their knowledge remains static and bounded by training data cutoffs. This creates an ever-widening gap with the rapidly evolving information landscape. RAG systems address this critical limitation by enabling LLMs to retrieve and incorporate external knowledge sources during inference. This transforms them from static repositories into adaptive, knowledge-aware systems.

The Multimodal Reality: Beyond Text-Only RAG. Current RAG systems face a critical limitation that severely restricts their real-world deployment. Existing frameworks operate under the restrictive assumption that knowledge corpus consists exclusively of plain textual documents. This assumption fundamentally misaligns with how information exists in authentic environments. Real-world knowledge repositories are inherently **heterogeneous and multimodal**, containing rich combinations of textual content, visual elements, structured data, and mathematical expressions. These diverse knowledge sources span multiple document formats and presentation mediums, from research papers and technical slides to web pages and interactive documents.

2.1.1 MOTIVATING RAG-ANYTHING

This multimodal reality introduces fundamental technical challenges that expose the inadequacy of current text-only RAG approaches. Effective multimodal RAG requires unified indexing strategies that can handle disparate data types, cross-modal retrieval mechanisms that preserve semantic relationships across modalities, and sophisticated synthesis techniques that can coherently integrate diverse information sources. These challenges demand a fundamentally different architectural approach rather than incremental improvements to existing systems.

The RAG-Anything framework introduces a unified approach for retrieving and processing knowledge from heterogeneous multimodal information sources. Our system addresses the fundamental challenge of handling diverse data modalities and document formats within a retrieval pipeline. The framework comprises three core components: universal indexing for multimodal knowledge, cross-modal adaptive retrieval, and knowledge-enhanced response generation. This integrated design enables effective knowledge utilization across modalities while maintaining computational efficiency.

2.2 UNIVERSAL REPRESENTATION FOR HETEROGENEOUS KNOWLEDGE

A key requirement for universal knowledge access is the ability to represent heterogeneous multimodal content in a unified, retrieval-oriented abstraction. Unlike existing pipelines that simply parse documents into text segments, RAG-Anything introduces **Multimodal Knowledge Unification**. This process decomposes raw inputs into atomic knowledge units while preserving their structural context and semantic alignment. For instance, RAG-Anything ensures that figures remain grounded in their captions, equations remain linked to surrounding definitions, and tables stay connected to explanatory narratives. This transforms heterogeneous files into a coherent substrate for cross-modal retrieval.

Formally, each knowledge source $k_i \in \mathcal{K}$ (e.g., a web page) is decomposed into atomic content units:

$$k_i \xrightarrow{\text{Decompose}} \{c_j = (t_j, x_j)\}_{j=1}^{n_i}, \quad (1)$$

where each unit c_j consists of a modality type $t_j \in \text{text, image, table, equation, } \dots$ and its corresponding raw content x_j . The content x_j represents the extracted information from the original knowledge source, processed in a modality-aware manner to preserve semantic integrity.